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BMJ Open

A systematic review and meta-analysis of studies examining the relationship between reported racism and health and wellbeing for children and adolescents

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CHILD HEALTH REVIEW PROTOCOL DRAFT

A SYSTEMATIC REVIEW AND META-ANALYSIS OF STUDIES EXAMINING THE RELATIONSHIP BETWEEN REPORTED RACISM AND HEALTH AND WELLBEING FOR CHILDREN AND ADOLESCENTS.

Naomi Priest[1, 2 *], Kate Doery[1, 2], Mandy Truong[3], Shuaijun Guo[2,4], Brigid Trenerry[5], Saffron Karlsen[6], Yvonne Kelly[7], Yin Paradies[8]

1. Australian National University, Centre for Social Research and Methods, Canberra, Australia
2. Murdoch Children's Research Centre, Population Health, Melbourne, Australia
3. Monash University, School of Nursing and Midwifery, Melbourne, Australia
4. University of Melbourne, Department of Pediatrics, Melbourne, Australia.
5. Singapore University of Technology and Design, Lee Kuan Yew Centre for Innovative Cities, Singapore
6. University of Bristol, School of Sociology, Politics and International Studies, Bristol, United Kingdom
7. University College London, Department of Epidemiology and Public Health, London, United Kingdom
8. Deakin University, School of Humanities and Social Science, Melbourne, Australia

* Corresponding Author Associate Professor Naomi Priest, The Australian National University, Canberra ACT 2600 Australia. Email: naomi.priest@anu.edu.au Phone: +61 2 6125 4849.

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ABSTRACT

Introduction: There is a growing body of research showing associations between experiences of racism and poor health and wellbeing outcomes for children and adolescents. The aim of this review protocol is to update the first systematic review conducted by Priest et al. 2013, including a meta-analysis of findings. Based on previous empirical data, it is anticipated that child and adolescent health will be negatively impacted by racism. This review will provide updated evidence of effect sizes across outcomes and identify moderators and mediators of relationships.

Methods and analysis: This systematic review and meta-analysis will include studies that explore associations between experiences of racism and racial dissemination with health outcomes of children and adolescents aged 0-24 years of age from any racial/ethnic/cultural group. Outcome measures include general health and wellbeing, physical health, mental health, healthcare utilisation and health behaviours. Exposure measures include self-reported and proxy reported personal experiences of racism and reported experiences of vicarious racism. The authors will conduct a comprehensive search of studies from the earliest time available to September 2020. All relevant studies will be screened with data extraction, quality appraisal and publication bias conducted independently by at least two authors.

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Ethics and Dissemination: This review will provide evidence for future research within the field and help to support policy and practice development. Results from this systematic review and meta-analysis will be widely disseminated to both academic and non-academic audiences.

Ethics approval is not required as this is a review of existing empirical findings.

Key Words: Protocol, Systematic-Review, Meta-analysis, Racism, Child Health.

Article Summary

- This is an updated systematic review which aims to update the findings from the first international review conducted by Priest et al. 2013. However, it is the first meta-analysis to be conducted exploring the relationship between racism and health in child and adolescent from all ethnic/racial/cultural backgrounds.
- This systematic review will show the health effects of racial discrimination on child and adolescent health, the key pathways by which racial discrimination influences these outcomes and identify potential moderators and mediators.
- Findings from this systematic review and meta-analysis will be used to provide recommendations for future research and inform the development of effective evidence-based strategies for addressing racism and ameliorating its harmful effects.
- This systematic review has a bias towards papers published in English as this review will only search studies published in English, meaning that studies not-published in English will not be included in this review. By doing so this review may not include all findings of all relevant studies.

INTRODUCTION

Racism and racial discrimination are widely recognised as critical determinants of health and health inequities for children and adolescents across populations and contexts[1-3]. Racism is a system of oppression that categorises and stratifies social groups into 'races', devalues and disadvantages those considered inferior and differentially allocates to them valued societal resources and opportunities[4 5]. Racism is expressed across multiple levels, including systemic and inter- and intra-personal levels, and operates in many forms including vicarious racism, whereby individuals experience racism on a secondary level, witnessing or being informed of family, friends and strangers experiencing racism[6 7]. Racial discrimination is the behavioural expression of racism and also manifests at systemic and individual levels. Currently the world is currently experiencing two global pandemics, COVID-

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19 and racism, with many people experiencing heightened levels of discrimination which is having negative impacts on people' health and wellbeing[8].

Research on racism and health has predominantly focused on interpersonal experiences, with considerable evidence documenting negative health effects across multiple outcomes[5 9 10]. However, most of this evidence remains focusing on adults, with far less research conducted among children and adolescents.

Priest et al.[1] conducted the first international systematic review of epidemiological studies on reported racial discrimination and the health and wellbeing of children and adolescents, including 121 studies. However since this report was published in 2013 the contribution of racism as a social determinant of health and wellbeing among children and adolescents has received growing attention[3]. There is increasing evidence of the impact of racism on pathophysiological processes (e.g. allostatic load and stress neurobiology) and biological markers (e.g. C-reactive protein and cortisol)[11] as well as on sleep[12-14] among children and adolescents.

A recent review of vicarious racism and child health found 30 studies published up to May 2016 compared with 10 studies in the previous 2013 review (with studies searched up to November 2011)[6]. This represents a three-fold increase in studies examining vicarious racism and child health in approximately four and a half years. Additionally, our original review found that two-thirds of the included studies were published between 2005-2012[1]. In light of the burgeoning research in the field, there is a need to review and reflect on the current evidence to inform future scholarship in this area.

This present systematic review and meta-analysis aims to update findings from the 2013 review conducted by Priest et al.[1] An updated systematic review is necessary to include new data, new methods, and updated analysis to a similar research question.[15] In this instance an updated systematic review is necessary due to changing social policy and demographic contexts and new health priorities globally, as well as an increase in the number of recent publications in this area, including in different country and population contexts. The first systematic review identified that there were a limited number of longitudinal studies that have explored the health effects of racism on children and a need to expand research in this area, with a focus needed on the complex pathways to which child and youth health is impacted by experiences of racial discrimination[1]. Priest et al. called for an increase in high-quality longitudinal research utilising robust multidimensional measures of racial discrimination[1]. As highlighted since this review was published in 2013, there has been a large increase in the amount of research being conducted in this field.

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This present systematic review and meta-analysis will use the previous review as a guide, building upon it and utilising an updated inclusion and exclusion strategy as well as expanding it to include a meta-analysis. As indicated by Garner et al.[15] an updated systematic review can have an updated inclusion criteria whilst answering a similar question.

The broad aims of this present systematic review and meta-analysis are:

- To further understanding of the health effects of racial discrimination on child and adolescent health, the key pathways by which racial discrimination influences these outcomes and identify potential moderators and mediators.
- To provide key recommendations for future research and inform the development of effective evidence-based strategies for addressing racism and ameliorating its harmful effects.

METHODS AND ANALYSIS

This systemic review and meta-analysis will follow the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)[16] with the PRISMA Protocols (PRISMA-P)[17] checklist followed for the writing of this protocol and is included as appendix 1.

Progress on this systematic review and meta-analysis will be updated on the International Prospective Register of Systematic Reviews (PROSPERO) to maintain transparency.

Inclusion criteria

Studies using quantitative methods including cross-sectional; prospective and retrospective cohort; and case–control designs will be included. Peer-reviewed journal articles (published or available as pre-prints), and dissertations/theses will be included. We will also include grey literature including published reports. Studies that do not report empirical associations between racism and child and adolescent health will not be included.

Participants

Participants will include children and adolescents aged up to 24 years from any racial/ethnic/cultural groups. The age range of participants has been updated since the previous review (which included participants up to 18 years) due to a shift in the definition of the age of adolescence.[18]

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Exposure

This review will include studies which focus on reported childhood experiences of racism as the exposure, including: self-reported racism, proxy reports from child's experiences or racism as reported by parents or carers including experiences of vicarious racism (for example, witnessing racism experienced by family or friends). There will be no restrictions placed on the timeframe of exposure to racism prior to the measurement. Retrospective adult population studies that report on childhood experiences of racism will be noted, but will not be included in our analysis.

Outcome measures

Studies will be considered if they measure health outcomes in children and adolescents. Health and wellbeing outcomes include measures of ill-health and illness as well as positive health outcomes across physical, mental and behavioural domains. As guided by previous reviews and research[1 9 12 19-24], the following health and wellbeing outcomes will be included:

1. Pregnancy and birth outcomes (e.g. premature birth, low birth weight)
2. General health and wellbeing
3. Physical health (infectious disease and chronic conditions and markers e.g. body mass index, waist hip ratio, blood pressure, metabolic and cardiovascular disease, overweight, obesity)
4. Mental health (e.g. social and emotional difficulties, psychological adjustment and distress, self-esteem, mental illness, suicide risk, sleep difficulties, psychosis)
5. Wellbeing, life satisfaction, quality of life, resilience
6. Positive mental health (e.g. self-esteem, self-worth and resilience)
7. Learning and developmental difficulties (e.g. developmental delay, concerns about learning, externalising behaviours such as violence etc., poor attachment etc.)
8. Health/risk behaviours (e.g. alcohol, tobacco, substance use)
9. Health care utilisation, healthcare costs, satisfaction with child health care system (use of screening tests, maternal child health care, access to health care and treatment, adherence to treatment)
10. Biological markers (e.g. inflammation and cardiometabolic markers)

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Exclusion criteria

Studies reporting the effects of reported racism on other outcomes (e.g. education, employment) will not be included. Studies where specific effects of racism cannot be isolated from broader measure of discrimination will be noted but will not be included in the meta-analysis. Only studies published in English will be included.

Qualitative studies or studies only reporting the prevalence of racism without identifying associations with health and wellbeing outcomes will not be included.

Data extraction and management

Search Strategy

The search strategy will be conducted in English and include studies from the earliest time available to September 2020. The search strategy will not be restricted to papers only published since the completion of the previous search strategy in 2011 as databases regularly back index studies and therefore some studies may have been missed by the original review. The search will be checked against the original search results to ensure that all studies that have been back indexed are also included.

The search will be conducted in the Ovid Medline, Ovid PsycInfo, PubMed, ERIC and ProQuest (for dissertation/theses) databases. Reference lists will also be hand-searched for relevant studies. The authors will also search google scholar and key websites as well as contacting experts within the field to make sure all relevant studies including grey literature are included.

The search will be performed using a string template combining search terms relevant to our study population, exposure and outcomes. The search strategy template has been developed in consultation with medical library staff utilising the previous search strategy by Priest et al.[1] as a template. The search template to be used for Medline is included as appendix 2, which will be updated accordingly for each database.

Selection of studies

One member of the review team will conduct the initial search in the selected databases with the search results to be imported into Endnote X9[25], with duplicates and papers not in English to be deleted. All titles and abstracts of studies identified in the search will be independently screened for eligibility for inclusion by members of the review team using Covidence. Full text studies will be assessed for final inclusion. Any discrepancies between members of the review team will be resolved by having a third member of the review team adjudicate the decision.

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Rationale for exclusion of studies will be noted throughout the screening process with a PRISMA flowchart[16] being used to show the full selection process of studies.

Further, studies with inappropriate and/or insufficient data to allow meta-analysis will be documented, but excluded from the quantitative analysis.

Data Extraction

Once the full text studies have been identified, members of the review team will extract the data using Airtable[26]. An independent member of the review team will compare and check for inconsistencies with discrepancies resolved through discussion. Data from some studies may appear in multiple publications. If publications include unique combinations of exposure and outcome variables they will be extracted as distinct data sets; meaning that one study may be included in the meta-analysis multiple times as different data-sets due to its use of multiple measures of health or racism.

Specifically, this review will examine the key characteristics of studies of reported racism and health and wellbeing among children and adolescents including:

- where and when studies have been conducted, the racial/cultural/ethnic background, age and gender of study populations, study designs, sample sizes, and data sources used;
- how reported racism is defined, how this exposure is measured in terms of method of administration, content and time frames of exposure, targets and perpetrators of racism, reactions/responses to racism and settings in which racism is experienced; and
- the prevalence of reported racism and direction/magnitude/effect sizes of associations found between reported racism and health across health outcomes, study and exposure characteristics and identify effect moderator and mediators of these associations.

Data to be extracted will include: authors; year of publication; study aims; study design (including sampling methods); definition of racism exposure measure (s) (including tools/instruments and psychometric properties when applicable; method of administration including informant(s); content and time frames of exposure, targets and perpetrators of racism, reactions/responses to racism and settings in which racism is experienced); health outcome measures; measure of racial/cultural/ethnic background; study location (country/region); place of residence (urban/rural), sample size; participant demographics (age, racial/cultural backgrounds, gender, religion, education, socio-economic status, migration status); study findings; prevalence of self-reported racism including

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exposure characteristics; nature of associations between self-reported racism and health and wellbeing outcomes (including subgroup analysis when reported (mean, SD, effect size)); reactions and responses to racism; confounders, effect moderator and mediators of these associations; study recommendations; and study quality/critical appraisal.

Effect sizes such as coefficients and p-values for each health outcome will be extracted. Both unadjusted and adjusted effect sizes will be extracted when available, and covariates included in models recorded. Only effect sizes for children and adolescents will be extracted. Where an overall effect size is reported across a range of ages, we will extract subgroup effect sizes when reported.

Assessment of study quality and bias

Studies included in the review will be critically appraised to determine the validity of the study's findings from the known literature and to provide readers with the ability to make an informed decision on the quality of these findings. The review team will use a variety of Critical Appraisal Tools, utilising tools that are most relevant to the methods used in the study. We will utilise the Systematic Review Toolbox to assist with finding the most appropriate tools.[27] These tools will include the Newcastle- Ottawa Scale[28] and the Joanna Briggs critical appraisal tools appropriate to relevant study designs. Any discrepancies between quality ratings will be resolved through discussion between members of the review team. The quality ratings will be used as a guide for data analysis and meta-analysis.

The review team will also assess the publication bias of the included studies. The review team will use a variety of measures to assess the publication bias of the included studies, these will include using the Risk Of Bias In Non-randomized Studies - of Exposures (ROBINS-E) Tool[29], an assessment tool under development which assesses seven different domains of bias in exposure studies,[30] as included as Appendix 3.

Analysis

Data that meet all inclusion criteria will first be summarised descriptively and then analysed statistically. Data analysis will be conducted using Airtable[26].

A descriptive summary with data tables will be produced to summarise the literature. Study characteristics will be presented in summary tables across key variables (including their specific design, study details, sample size, age and racial or ethnic background of participants, exposure and outcome measures included).

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Although meta-analysis is planned this will only become apparent when extracted data are reviewed for feasibility. If data is available we will conduct analyses of associations between racism and health for different health outcome measures, and at different time points. If possible, we will use random-effects models to aggregate effect sizes. Subgroup analyses will be conducted for age, gender and ethnicity if possible. To assess the heterogeneity of studies, we will use the Q-statistic test and the I^2 statistic. If the test for heterogeneity denoted as I^2 (if $I^2 \leq 25\%$), studies will be considered homogeneous.

DISCUSSION

As this is an updated systematic review and meta-analysis, we expect that whilst there has been a significant amount of recent research conducted in this space, we do not anticipate our findings to be vastly different from our original review. This review and meta-analysis will incorporate studies with participants from all ethnic/racial/cultural backgrounds, and studies will not be limited to any one country or geographic area, and in doing so we anticipate to show that this is a problem facing not just one specific population but children globally. That is, we expect the review to show that racism and racial discrimination negatively impact on multiple health outcomes in children and adolescents from different ethnic/racial/cultural backgrounds and across contexts. We expect an increase of research in outcomes not considered in the original review, including sleep and inflammatory and immune biomarkers, as well as markers of epigenetic risk and cellular aging and of endocrine and hormonal function. Increased attention on younger age groups, vicarious as well as direct exposure, longitudinal associations, and populations and settings outside of the United States of America is also anticipated.

The world is currently facing two global pandemics, with high levels of racial discrimination being experienced globally throughout the COVID-19 pandemic[8], and racism has been drawing increased media attention. Exploring the health effects of racism and discrimination is paramount. Due to the expectant increase in research surrounding this topic, a key contribution of the current study is to conduct a meta-analysis, which was not able to be conducted before. We expect that through this meta-analysis we will be able to show rigorous and robust evidence showing the relationship between experiences of racism and health and wellbeing outcomes for children and adolescents. As this is the first meta-analysis of these studies, it will provide an evidence base for future research exploring the effect of racism and child health, as well as for policy development and service delivery.

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ETHICS AND DISSEMINATION

Ethics approval is not required as this is a review of existing empirical findings and no primary data will be collected throughout the research.

The results from this review will be disseminated in peer-review publications, conference presentations as well as communicated more broadly through factsheets and summaries disseminated through academic institution press release and policy and practice partners.

Progress on this systematic review and meta-analysis will be updated on the International Prospective Register of Systematic Reviews (PROSPERO) (PROSPERO ID 184055) to maintain transparency.

AUTHOR CONTRIBUTIONS

NP conceptualised the review and contributed to all aspects of the protocol. KD drafted the protocol, developed the search strategy with medical librarians. MT, JG, BT, SK, YK and YP reviewed the protocol draft.

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COMPETING INTERESTS

The authors have no conflicts of interest relevant to this article to disclose.

REGISTRATION

The systematic review will be registered with PROSPERO, ID number 184055.

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Appendix 1- PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review
Update	1b	If the protocol is for an update of a previous systematic review, identify as such
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments
Support:		
Sources	5a	Indicate sources of financial or other support for the review
Sponsor	5b	Provide name for the review funder and/or sponsor
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review

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Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Appendix 2- Search Strategy

Search date: 18/7/2020

Database(s): Ovid MEDLINE(R) ALL 1946 to July 16, 2020

Search Strategy:

1	Prejudice/ or Racism/
2	(racism or racial-discriminat* or racial-prejudice or racist-event* or racist-episode* or racial-stereotype* or race-related-stress).tw,kf.
3	((discriminat* or bias* or prejudic* or hostil* or harass* or bully* or cyberbull* or cyber-bull* or (unfair* adj1 treat*) or oppress*) adj3 (race or racial* or ethnic* or cultur* or religio* or migrant* or refugee* or asylum)).tw,kf.
4	(newborn* or new-born* or baby or babies or neonat* or neo-nat* or infan* or toddler* or pre-schooler* or preschooler* or kinder or kinders or kindergarten* or boy or boys or girl or girls or child or children or childhood or pediatric* or paediatric* or adolescen* or youth or youths or teen or teens or teenage* or school-age* or schoolage* or school-child* or schoolchild* or school-girl* or schoolgirl* or school-boy* or schoolboy* or young-person* or young-people).af.
5	Child Welfare/ or pediatric obesity/et, ep, pc
6	(Prejudice/ or *Racism/ or 2 or 3) and 5
7	obesity/et, ep, pc or body mass index/ or overweight/pc
8	Waist-Hip Ratio/
9	Blood Pressure/ or Biomarkers/
10	Hypertension/et, ep, pc
11	exp Cardiovascular Diseases/et, ep, pc
12	depression/et, ep, pc or anxiety/et, ep, pc
13	Mental Health/
14	Stress, Psychological/et, ep, pc
15	Sleep/
16	exp Sleep Wake Disorders/et, ep, pc
17	"Quality of Life"/
18	Resilience, Psychological/ or exp adaptation, psychological/
19	exp substance-related disorders/et, ep, pc or alcohol-related disorders/et, ep, pc
20	smoking/et, ep or exp tobacco smoking/et, ep, pc
21	Mental Disorders/et, ep, pc
22	Self Concept/
23	personal satisfaction/
24	exp suicide/et, ep, pc
25	conduct disorder/et, ep, pc or aggression/et, ep, pc
26	pregnancy outcome/
27	(health-care or healthcare or health-service* or clinic? or ill-health or wellbeing or well-being or disease* or illness* or bmi or body-mass-index or anthropometric* or WHR or waist-hip-ratio or hypertension or blood-pressure or cardiometabolic or cardio-metabolic or biomarker* or obese or obesity or overweight or depress* or anxiety or anxious* or mental-health or mental-disorder* or stress or distress* or suicid* or sleep or psychosis or tobacco or smoke* or smoking or drug? or alcohol* or substance-use or substance-related-disorder* or resilien* or self-esteem or self-worth or self-concept or quality-of-life or life-satisfaction or personal-satisfaction or conduct-disorder* or aggression).tw,kf.
28	((social or behavio* or emotion* or developmental* or psychological* or learning*) adj3 (difficul* or problem* or delay* or adjust*)).tw,kf.
29	(((pregnancy or birth or gestation*) and (outcome* or preterm or pre-term or premature or small-for-gestational-age)) or low-birthweight or low-birth-weight).tw,kf.
30	7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26
31	*obesity/et, ep, pc or *body mass index/ or *overweight/pc or *Waist-Hip Ratio/ or (*Blood Pressure/ or *Biomarkers/) or *Hypertension/et, ep, pc or exp *Cardiovascular Diseases/et, ep, pc or (*depression/et, ep, pc or *anxiety/et, ep, pc) or *Mental Health/ or *Stress, Psychological/et, ep, pc or *Sleep/ or exp *Sleep Wake Disorders/et, ep, pc or *"Quality of Life"/ or (*Resilience, Psychological/ or exp *adaptation, psychological/) or (exp *substance-related disorders/et, ep, pc or *alcohol-related disorders/et, ep, pc) or (*smoking/et, ep or exp *tobacco smoking/et, ep, pc) or *Mental Disorders/et, ep, pc or *Self Concept/ or *personal satisfaction/ or exp *suicide/et, ep, pc or (*conduct disorder/et, ep, pc or *aggression/et, ep, pc) or *pregnancy outcome/
32	(Prejudice/ or *Racism/ or 2 or 3) and (27 or 28 or 29 or 31) and 4

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33	6 or 32
34	limit 33 to (comment or editorial or letter)
35	**33 not 34**

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Appendix 3: The ROBINS-E tool (Risk Of Bias In Non-randomized Studies - of Exposures)

Preliminary tool for risk of bias in exposure studies (1): At protocol stage

Specify the research question by defining a generic target experiment

Participants	
Experimental exposure	
Control exposure	

List the confounding domains relevant to all or most studies

List the possible co-exposures that could differ between exposure groups and could have an impact on study outcomes

List the criteria used to determine the accuracy of exposure measurement

Factors to consider when evaluating health outcome assessment

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Preliminary tool for risk of bias in exposure studies (2): For each study

Specify a target experiment specific to the study.

☐ The protocol-specified target experiment fully applies

OR

Participant	
Experimental exposure	
Control exposure	

Specify the outcome

Specify which outcome is being assessed for risk of bias (typically from among those earmarked for the Summary of Findings table). Specify whether this is a proposed benefit or harm of exposure.

Is your aim for this study…?

- ☐ to assess the effect of initiating intervention (as in an intention-to-treat analysis)
- ☐ to assess the effect of initiating and adhering to intervention (as in a per-protocol analysis)
- ☐ other (specify)

Specify the numerical result being assessed

In case of multiple alternative analyses being presented, specify the numeric result (e.g. RR = 1.52 (95% CI 0.83 to 2.77) and/or a reference (e.g. to a table, figure or paragraph) that uniquely defines the result being assessed.

Preliminary consideration of confounders

Complete a row for each important confounding area (i) listed in the review protocol; and (ii) relevant to the setting of this particular study, or which the study authors identified as potentially important.

"Important" confounding areas are those for which, in the context of this study, adjustment is expected to lead to a clinically important change in the estimated effect of the exposure. "Validity" refers to whether the confounding variable or variables fully measure the area, while "reliability" refers to the precision of the measurement (more measurement error means less reliability).

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(i) Confounding areas listed in the review protocol				
Confounding area	Measured variable(s)	Is there evidence that controlling for this variable was unnecessary?*	Is the confounding area measured validly and reliably by this variable (or these variables)?	OPTIONAL: Is adjusting for this variable (alone) expected to move the effect estimate up or down?
			Yes / No / No information	Favor intervention / Favor control / No information

(ii) **Additional confounding areas relevant to the setting of this particular study, or which the study authors identified as important**				
Confounding area	Measured variable(s)	Is there evidence that controlling for this variable was unnecessary?*	Is the confounding area measured validly and reliably by this variable (or these variables)?	OPTIONAL: Is adjusting for this variable (alone) expected to move the effect estimate up or down?
			Yes / No / No information	Favor intervention / Favor control / No information

* In the context of a particular study, variables can be demonstrated not to be confounders and so not included in the analysis: (a) if they are not predictive of the outcome; (b) if they are not predictive of exposure; or (c) because adjustment makes no or minimal difference to the estimated effect of the primary parameter. Note that "no statistically significant association" is not the same as "not predictive".

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Preliminary consideration of criteria used to determine the accuracy of measurement of exposure and outcome

Complete a row for each measure listed in the study for the (i) exposure and (ii) outcome. Of the measures listed in the protocol, consider the sensitivity, specificity, and confidence in the methods used in the study.

(i) Exposure measurement method listed in the study		
Method of measurement	Measured exposure	Is the exposure measured validly and reliably by this method (or these methods)?
		Yes / No / No information

(ii) Outcome measurement method listed in the study		
Method of measurement	Measured outcome	Is the outcome measured validly and reliably by this method (or these methods)?
		Yes / No / No information

Preliminary consideration of co-exposures

Complete a row for each important co-intervention (i) listed in the review protocol; and (ii) relevant to the setting of this particular study, or which the study authors identified as important.
"Important" co-interventions are those for which, in the context of this study, adjustment is expected to lead to a clinically important change in the estimated effect of the intervention.

(i) Co-exposures listed in the review protocol		
Co-exposure	Is there evidence that controlling for this co-exposure was unnecessary (e.g., because it was not administered)?	Is presence of this co-exposure likely to favor outcomes in the experimental or the control group
		Favor experimental / Favor comparator / No information

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		Favor experimental / Favor comparator / No information
		Favor experimental / Favor comparator / No information

(ii) Additional co-exposures relevant to the setting of this particular study, or which the study authors identified as important		
Co-exposure	Is there evidence that controlling for this co-exposure was unnecessary (e.g., because it was not administered)?	Is presence of this co-exposure likely to favor outcomes in the experimental or the control group
		Favor experimental / Favor comparator / No information
		Favor experimental / Favor comparator / No information
		Favor experimental / Favor comparator / No information

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Risk of bias assessment (cohort-type studies)

Bias due to confounding	1.1 **Is there potential for confounding of the effect of exposure in this study? If N or PN to 1.1:** the study can be considered to be at low risk of bias due to confounding and no further signaling questions need be considered	Y / PY / PN / N	[Description]
	If Y/PY to 1.1, answer 2.1 and 1.3 to determine whether there is a need to assess time-varying confounding:		
	1.2. **If Y or PY to 1.1:** Was the analysis based on splitting follow up time according to exposure received? **If N or PN to 1.2**, answer questions 1.4 to 1.6, which relate to baseline confounding	NA / Y / PY / PN / N / NI	[Description]
	1.3. **If Y or PY to 1.2**: Were exposure discontinuations or switches likely to be related to factors that are prognostic for the outcome?	NA / Y / PY / PN / N / NI	[Description]
	If N or PN to 1.3, answer questions 1.4 to 1.6, which relate to baseline confounding		
	1.4. Did the authors use an appropriate analysis method that adjusted for all the critically important confounding areas?	NA / Y / PY / PN / N / NI	[Description]
	1.5. **If Y or PY to 1.4**: Were confounding areas that were adjusted for measured validly and reliably by the variables available in this study?	NA / Y / PY / PN / N / NI	[Description]
	1.6. Did the authors avoid adjusting for post-exposure variables?	NA / Y / PY / PN / N / NI	[Description]
	If Y or PY to 1.3, answer questions 1.7 and 1.8, which relate to time-varying confounding		
	1.7. Did the authors use an appropriate analysis method that adjusted for all the critically important confounding areas and for time-varying confounding?	NA / Y / PY / PN / N / NI	[Description]

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	1.8. **If Y or PY to 1.7**: Were confounding areas that were adjusted for measured validly and reliably by the variables available in this study?	NA / Y / PY / PN / N / NI	[Description]
	Risk of bias judgement	Low / Moderate / Serious / Critical / NI	[Support for judgement]
	Optional: What is the predicted direction of bias due to confounding?	Favors experimental / Favors comparator / Unpredictable	[Rationale]
Bias in selection of participants into the study	2.1. Was selection of participants into the study (or into the analysis) based on variables measured after the start of the exposure? **If N or PN to 2.1 go to 2.4**	Y / PY / PN / N / NI	[Description]
	2.2. **If Y/PY to 2.1:** Were the post-exposure variables that influenced selection associated with exposure?	Y / PY / PN / N / NI	[Description]
	2.3. **If Y/PY to 2.2:** Were the post-exposure variables that influenced eligibility selection influenced by the outcome or a cause of the outcome?	NA / Y / PY / PN / N / NI	[Description]
	2.4 Do start of follow-up and start of exposure coincide for most participants?	NA / Y / PY / PN / N / NI	[Description]
	2.5 If Y/PY to 2.2 and 2.3, or N/PN to 2.4: Were adjustment techniques used that are likely to correct for the presence of selection biases?	NA / Y / PY / PN / N / NI	[Description]
	Risk of bias judgement	Low / Moderate / Serious / Critical / NI	[Support for judgement]
	Optional: What is the predicted direction of bias due to selection of participants into the study?	Favors experimental / Favors comparator / Towards null /Away from null / Unpredictable	[Rationale]
Bias in classification of exposures	3.1 Is exposure status well defined?	Y / PY / PN / N / NI	[Description]
	3.2 Did entry into the study begin with start of the exposure?	Y / PY / PN / N / NI	[Description]
	3.3 Was information used to define exposure status recorded prior to outcome assessment?	Y / PY / PN / N / NI	[Description]
	3.4 Could classification of exposure status have been affected by knowledge of the outcome or risk of the outcome?	Y / PY / PN / N / NI	[Description]

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	3.5 Were exposure assessment methods robust (including methods used to input data)?	Y / PY / PN / N / NI	[Description]
	Risk of bias judgement	Low / Moderate / Serious / Critical / NI	[Support for judgement]
	Optional: What is the predicted direction of bias due to measurement of outcomes or exposures?	Favors experimental / Favors comparator / Towards null /Away from null / Unpredictable	[Rationale]
Bias due to departures from intended exposures	4.1. Is there concern that changes in exposure status occurred among participants? **If your aim for this study is to assess the effect of initiating and adhering to an exposure (as in a per-protocol analysis), answer questions 4.2 and 4.3, otherwise continue to 4.4 if Y or PY to 4.1.**	Y / PY / PN / N / NI	[Description]
	4.2. Did many participants switch to other exposures?	Y / PY / PN / N / NI	[Description]
	4.3. Were the critical co-exposures balanced across exposure groups?	Y / PY / PN / N / NI	[Description]
	4.4. **If NY/PN PY to 4.1, or Y/PY to 4.2, or 4.3:** Were adjustment techniques used that are likely to correct for these issues?	NA / Y / PY / PN / N / NI	[Description]
Risk of bias judgement	Low / Moderate / Serious / Critical / NI	[Support for judgement]	
Optional: What is the predicted direction of bias due to departures from the intended exposures?	Favors experimental / Favors comparator / Towards null /Away from null / Unpredictable	[Rationale]	
Bias due to missing data	5.1 Were there missing outcome data?	Y / PY / PN / N / NI	[Description]
	5.2 Were participants excluded due to missing data on exposure status?	Y / PY / PN / N / NI	[Description]
	5.3 Were participants excluded due to missing data on other variables needed for the analysis?	Y / PY / PN / N / NI	[Description]
	5.4 **If Y/PY to 5.1, 5.2 or 5.3:** Are the proportion of participants and reasons for missing data similar across exposures?	NA / Y / PY / PN / N / NI	[Description]
	5.5 **If Y/PY to 5.1, 5.2 or 5.3:** Were appropriate statistical methods used to account for missing data?	NA / Y / PY / PN / N / NI	[Description]

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	Risk of bias judgement	Low / Moderate / Serious / Critical / NI	[Support for judgement]
	Optional: What is the predicted direction of bias due to missing data?	Favors experimental / Favors comparator / Towards null /Away from null / Unpredictable	[Rationale]
Bias in measurement of outcomes	6.1 Could the outcome measure have been influenced by knowledge of the exposure received?	Y / PY / PN / N / NI	[Description]
	6.2 Was the outcome measure sensitive?	Y / PY / PN / N / NI	[Description]
	6.3 Were outcome assessors unaware of the exposure received by study participants?	Y / PY / PN / N / NI	[Description]
	6.4 Were the methods of outcome assessment comparable across exposure groups?	Y / PY / PN / N / NI	[Description]
	6.5 Were any systematic errors in measurement of the outcome unrelated to exposure received?	Y / PY / PN / N / NI	[Description]
	Risk of bias judgement	Low / Moderate / Serious / Critical / NI	[Support for judgement]
	Optional: What is the predicted direction of bias due to measurement of outcomes?	Favors experimental / Favors comparator / Towards null /Away from null / Unpredictable	[Rationale]
Bias in selection of	Is the reported effect estimate likely to be selected, on the basis of the results, from...?		
the reported result	7.1. ... multiple outcome *measurements* within the outcome domain?	Y / PY / PN / N / NI	[Description]
	7.2 ... multiple *analyses* of the exposure-outcome relationship?	Y / PY / PN / N / NI	[Description]
	7.3 ... different *subgroups*?	Y / PY / PN / N / NI	[Description]
	Risk of bias judgement	Low / Moderate / Serious / Critical / NI	[Support for judgement]

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	Optional: What is the predicted direction of bias due to selection of the reported result?	Favors experimental / Favors comparator / Towards null /Away from null / Unpredictable	[Rationale]
Overall bias	**Risk of bias judgement**	Low / Moderate / Serious / Critical / NI	[Support for judgement]
	Optional: What is the overall predicted direction of bias for this outcome?	Favors experimental / Favors comparator / Towards null /Away from null / Unpredictable	[Rationale]

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BMJ Open

An updated systematic review and meta-analysis of studies examining the relationship between reported racism and health and wellbeing for children and youth: A protocol

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Complete List of Authors:	Priest, Naomi; Australian National University College of Arts and Social Sciences, Centre for Social Research and Methods; Murdoch Childrens Research Institute, Population Health Doery, Kate; Australian National University College of Arts and Social Sciences, Centre for Social Research and Methods; Murdoch Childrens Research Institute, Population Health Truong, Mandy; Monash University, School of Nursing and Midwifery Guo, Shuaijun; Murdoch Childrens Research Institute, Population Health; The University of Melbourne, Department of Pediatrics Perry, Ryan; Australian National University College of Arts and Social Sciences, Centre for Social Research and Methods; Murdoch Childrens Research Institute, Population Health Trenerry, Brigid; Singapore University of Technology and Design, Lee Kuan Yew Centre for Innovative Cities Karlsen, Saffron; University of Bristol, School of Sociology, Politics and International Studies Kelly, Yvonne; University College London, Epidemiology and Public Health Paradies, Yin; Deakin University Faculty of Arts and Education, School of Humanities and Social Science
Primary Subject Heading:	Epidemiology
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Keywords:	Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Community child health < PAEDIATRICS, PUBLIC HEALTH

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AN UPDATED SYSTEMATIC REVIEW AND META-ANALYSIS OF STUDIES EXAMINING THE RELATIONSHIP BETWEEN REPORTED RACISM AND HEALTH AND WELLBEING FOR CHILDREN AND YOUTH: A PROTOCOL

Naomi Priest[1, 2 *], Kate Doery[1, 2], Mandy Truong[3], Shuaijun Guo[2,4], Ryan Perry[1, 2] Brigid Trenerry[5], Saffron Karlsen[6], Yvonne Kelly[7], Yin Paradies[8]

1. Australian National University, Centre for Social Research and Methods, Canberra, Australia
2. Murdoch Children's Research Institue, Population Health, Melbourne, Australia
3. Monash University, School of Nursing and Midwifery, Melbourne, Australia
4. University of Melbourne, Department of Pediatrics, Melbourne, Australia.
5. Singapore University of Technology and Design, Lee Kuan Yew Centre for Innovative Cities, Singapore
6. University of Bristol, School of Sociology, Politics and International Studies, Bristol, United Kingdom
7. University College London, Department of Epidemiology and Public Health, London, United Kingdom
8. Deakin University, School of Humanities and Social Science, Melbourne, Australia

* Corresponding Author Associate Professor Naomi Priest, Australian National University, Canberra ACT 2600 Australia. Email: naomi.priest@anu.edu.au Phone: +61 2 6125 4849.

Word Count: 3219

ABSTRACT

Introduction: Racism is a critical determinant of health and health inequities for children and youth. This protocol aims to update the first systematic review conducted by Priest et al. 2013, including a meta-analysis of findings. Based on previous empirical data, it is anticipated that child and youth health will be negatively impacted by racism. Findings from this review will provide updated evidence of effect sizes across outcomes and identify moderators and mediators of relationships between racism and health.

Methods and analysis: This systematic review and meta-analysis will include studies that examine associations between experiences of racism and racial discrimination with health outcomes of children and youth aged 0-24 years. Exposure measures include self-reported or proxy reported systemic, interpersonal and intrapersonal racism. Outcome measures include general health and wellbeing, physical health, mental health, biological markers, healthcare utilisation and health behaviours. A comprehensive search of studies from the earliest time available to October 2020 will be conducted. A random-effects meta-analysis will examine the average effect of racism on a range of health outcomes. Study-level moderation will test the difference in effect sizes with regard to various sample and exposure characteristics. This review has been registered with the International Prospective Register of Systematic Reviews (PROSPERO) ID CRD42020184055.

Ethics and Dissemination: This review will provide evidence for future research within the field and help to support policy and practice development. Results will be widely disseminated to both academic and non-academic audiences

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through peer review publications, community summaries, and presentations to research, policy, practice and community audiences.

Key Words: Protocol, Systematic-Review, Meta-analysis, Racism, Child Health.

Strengths and Limitations

- This protocol aims to update the international review of racism and child health conducted by Priest et al.2013
- The updated protocol now extends to include study participants 0-24 years to recognise the importance of youth development.
- This protocol includes a broad search strategy that aims to capture varied exposure measures of racism and racial discrimination as well as health and wellbeing outcomes.
- This protocol describes a meta-analysis to be conducted exploring relationships between racism and health among children and youth.
- This systematic review has a bias towards papers published in English, meaning that studies not-published in English will not be included in this review.

INTRODUCTION

Racism and racial discrimination are widely recognised as critical determinants of health and health inequities for children and youth across populations and contexts.[1-3] Racism is a system of oppression that categorises and stratifies social groups into 'races', devalues and disadvantages those considered inferior and differentially allocates to them valued societal resources and opportunities.[4 5] Racism is expressed across multiple levels, including systemic or structural racism, embedded within society; interpersonal racism between two or more individuals and includes racially motivated assaults or abuse; and intrapersonal racism, whereby people take on negative stereotypes and beliefs about themselves.[6-8] Racism and racial discrimination operates in many forms including direct and vicarious racism (second hand racism), whereby individuals experience racism on a secondary level, witnessing or being informed of family, friends and strangers experiencing racism.[9 10]

Research on racism and health has predominantly focused on interpersonal experiences, with considerable evidence documenting negative health effects across multiple outcomes, including physical and mental health outcomes.[5 11 12] However, most of this evidence focused on adults, with far less research conducted among children and youth.

Priest et al.[1] conducted the first international systematic review of quantitative studies on reported racial discrimination and the health and wellbeing of children and youth, including 121 studies. Since this report was published in 2013, the contribution of racism as a social determinant of health and wellbeing among children and youth has received growing

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attention.[3] There is increasing evidence of the impact of racism on pathophysiological processes (e.g. allostatic load and stress neurobiology) and biological markers (e.g. C-reactive protein and cortisol)[13] as well as on health behaviours such as sleep[14-16] among children and youth. The American Academy of Pediatrics recently issued a policy statement highlighting the impact that racism has on young people's health and health inequities and that addressing racism needs to be an urgent priority.[2]

A recent review of vicarious racism and child health found 30 studies published up to May 2016 compared with 10 studies in the previous 2013 review (with studies searched up to November 2011).[9] This represents a three-fold increase in studies examining vicarious racism and child health in approximately four and a half years. Additionally, our original review found that two-thirds of the included studies were published between 2005-2012.[1] In light of the growing research in the field, there is a need to review and reflect on the current evidence to inform future scholarship in this area.

This present systematic review and meta-analysis aims to update findings from the 2013 review conducted by Priest et al.[1] An updated systematic review is necessary to include new data, new methods and updated analysis.[17] In this instance an updated systematic review is necessary due to changing social policy and demographic contexts and new health priorities globally, as well as an increase in the number of recent publications in this area, including in different country and population contexts. The first systematic review identified that there were a limited number of longitudinal studies that have explored the health effects of racism on children and a need to expand research in this area, with a focus needed on the complex pathways to which child and youth health is impacted by experiences of racial discrimination.[1] Priest et al. called for an increase in high-quality longitudinal research utilising robust multidimensional measures of racial discrimination.[1] As highlighted since this review was published in 2013, there has been a large increase in the amount of research being conducted in this field.

To answer our research question "*To what extent are experiences of racism associated with health and wellbeing outcomes among children and youth compared to those who experience no or less racism?*" We will use the previous review (Priest et al., 2013) as a guide, building upon it and utilising an updated inclusion and exclusion strategy as well as expanding it to include a meta-analysis. As indicated by Garner et al.,[17] an updated systematic review can have an updated inclusion criteria whilst answering a similar question. This systematic review and meta-analysis aims to quantify the effects of racial discrimination on child and youth health; examine the key pathways by which racial discrimination influences these outcomes; and identify potential moderators and mediators between racism and health. This review will provide key recommendations for future research and inform the development of effective evidence-based strategies for addressing racism and ameliorating its harmful effects.

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METHODS AND ANALYSIS

This systematic review and meta-analysis will follow the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)[18] with the PRISMA Protocols (PRISMA-P)[19] checklist followed for the writing of this protocol (see supplementary file 1). Progress on this systematic review and meta-analysis will be updated on the International Prospective Register of Systematic Reviews (PROSPERO) to maintain transparency (PROSPERO ID CRD42020184055).

Inclusion criteria

This report will include primary empirical studies that use quantitative methods including but not limited to cross-sectional; prospective and retrospective cohort; case–control designs; quasi-experimental studies and randomised control trials. Peer-reviewed journal articles (published or available as pre-prints), and dissertations/theses will be included. We will also include grey literature such as published reports. Studies that do not report primary empirical associations between racism and child and youth health will not be included. Editorials and commentaries will not be included unless they report primary empirical data. According to the PICO (Population, Intervention, Comparison and Outcomes) tool outlined in the Cochrane Handbook,[20] we defined our population, intervention (exposure of interest) and specific outcome measures.

Population

Participants will include children and youth aged up to 24 years from any racial/ethnic/cultural groups. The age range of participants has been updated since the previous review (which included participants up to 18 years) to include children and youth as per the United Nations definition.[21-23] This broader age range will allow for consideration of the health impacts of racism into late adolescence which is now considered by the Lancet Commission on Adolescent Health as extending to 24 years.[23]

Exposure

This review will extract childhood experiences of racism and racial discrimination exposures, synthesised across three main categories in the meta-analysis: systemic racism; interpersonal experiences including direct and vicarious or second hand experiences (for example, witnessing or hearing about racism experienced by family, friends or a group an individual belongs to); and intrapersonal racism (for example, adapting racist attitudes and/or beliefs within their world views). Both personal and proxy reports (by parents or caregivers) of experiences of racism will be recorded, as will the source of racism (categorised as peers, teachers, and community).

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Terms used to determine the exposure include but are not limited to racism, discrimination, prejudice, harassment, bullying, stereotypes and unfair treatment where reason includes the victims' race/ethnicity/cultural background or proxy indicators such as migrant background, skin colour, language or accent. Although we recognise that religion is often highly racialised,[24] studies of religious discrimination will not be included as we consider this a related, but distinct form of discrimination. Religious discrimination is distinguished by being an assault on an individual or group's belief system and warrants independent investigation.[25]

There will be no restrictions placed on the timeframe of exposure to racism prior to the measurement. Retrospective adult population studies that report on childhood experiences of racism will be noted, but will not be included in our analysis.

Outcome measures

Studies will be considered if they measure health outcomes in children and youth. Health and wellbeing outcomes include measures of ill-health and illness as well as positive health outcomes across physical, mental and behavioural domains. As guided by previous reviews and research, [1 11 14 26-31] the following health and wellbeing outcomes will be included:

1. Pregnancy and birth outcomes (e.g. premature birth, low birth weight)
2. General health and wellbeing, life satisfaction and quality of life
3. Physical health (infectious disease and chronic conditions and markers e.g. body mass index, waist hip ratio, blood pressure, metabolic and cardiovascular disease, overweight, obesity)
4. Negative mental health (e.g. social and emotional difficulties, psychological distress, mental illness, suicide risk, self-harm, psychosis, antisocial behaviours including aggression and violence)
5. Positive mental health (e.g. self-esteem, self-worth and resilience)
6. Health behaviours (e.g. alcohol, tobacco, substance use) and sleep
7. Health care utilisation, healthcare costs, satisfaction with child health care system (use of screening tests, maternal child health care, access to health care and treatment, adherence to treatment)
8. Biological markers (e.g. inflammation and cardiometabolic markers)

Exclusion criteria

Studies reporting the effects of reported racism on other outcomes (e.g. cognitive development, education, employment) will not be included. Studies that do not attribute discrimination experiences to race or ethnicity but only report generalised discrimination or unfair treatment without attribution will not be included. Studies published in a language other than English will not be included. Qualitative studies or studies only reporting the prevalence of racism without identifying associations with health and wellbeing outcomes will not be included.

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Data extraction and management

Search Strategy

The search strategy will be conducted in English and include studies from the earliest time available to October 2020. The search strategy will not be restricted to papers only published since the completion of the previous search strategy in 2011 as databases regularly back index studies and therefore some studies may have been missed by the original review.[1] The search will be checked against the original search results to ensure that all studies that have been back indexed are also included.

The search will be conducted in the Ovid Medline, Ovid PsycInfo, PubMed, ERIC and ProQuest (for dissertation/theses) databases. Reference lists of included studies will also be hand-searched for additional relevant studies. The authors will also search google scholar to identify papers and reports citing the previous review and will search grey literature databases including Open Grey, OpenDOAR and New York Academy of Medicine using keywords from the search strategy.

The search will be performed using a string template combining search terms relevant to our study population, exposure and outcomes. The search strategy template has been developed in consultation with medical library staff utilising the previous search strategy by Priest et al.[1] as a template. The search template to be used for Medline is included as supplementary file 2, which will be updated accordingly for each database.

Selection of studies

One member of the review team will conduct the initial search in the selected databases with the search results to be imported into Endnote X9,[32] All titles and abstracts of studies identified in the search will be independently screened for eligibility for inclusion independently by two members of the review team using Covidence[33] with any discrepancies resolved by the lead author. Duplicates and papers not in English will be deleted and noted in the PRISMA flowchart.[18] Full text studies will be assessed for final inclusion. Any discrepancies between members of the review team will be resolved by having a third member of the review team adjudicate the decision. Rationale for exclusion of studies will be noted throughout the screening process with a PRISMA flowchart[18] being used to show the full selection process of studies.

Data Extraction

Once the full-text studies have been identified, members of the review team will extract the data using Airtable.[34] Two reviewers will independently extract data from each study, with inconsistencies and discrepancies resolved through discussion. Data from some studies may appear in multiple publications. If publications include unique combinations of

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exposure and outcome variables they will be extracted as distinct data sets; meaning that one study may be included in the meta-analysis multiple times as different data-sets due to its use of multiple measures of health or racism.

This review will examine the key characteristics of studies of reported racism and health and wellbeing among children and youth. Data to be extracted will include: authors; year of publication; study design (including sampling methods); definition of racism, exposure measure (s) (including tools/instruments and psychometric properties when applicable, method of administration including informant(s), content and time frames of exposure, targets and perpetrators, reactions/responses to racism, and settings in which racism is experienced); health and wellbeing outcome measures; measures of racial/cultural/ethnic background; study location (country/region); place of residence (urban/rural), sample size; participant demographics (age, racial/cultural backgrounds, gender, socio-economic status, migration status); study findings; prevalence of self-reported racism including exposure characteristics; nature of associations between self-reported racism and health and wellbeing outcomes including subgroup analysis when reported (mean, standard deviation, effect size); confounders, effect moderator and mediators of these associations; and study quality/critical appraisal.

Effect sizes such as coefficients and p-values for each health outcome will be extracted. Both unadjusted and adjusted effect sizes will be extracted when available, and covariates included in models recorded. Where an overall effect size is reported across a range of age groups, we will extract subgroup effect sizes when reported. In this case, only effect sizes for children and youth will be extracted.

Data including study characteristics, participant characteristics, and exposure and outcome characteristics will be extracted to be included in the narrative synthesis but will not be included in the meta-analysis.

Assessment of study quality and bias

Studies included in the review will be critically appraised to determine the validity of the study's findings from the known literature and to provide readers with the ability to make an informed decision on the quality of these findings. Two members of the review team will independently rate studies selected for the meta-analysis using the Newcastle-Ottawa Scale, a widely used tool for evaluating the quality of non-randomised studies on a range of criteria.[35] The Newcastle-Ottawa Scale (supplementary fiile 3) was determined to be the most appropriate instrument for quality assessment via a consideration of tools available in the Systematic Review Toolbox.[36] Any discrepancies between quality ratings will be resolved through discussion with the lead author. The quality ratings for each study will be provided in a table in the supplementary material and studies with scores deemed less than satisfactory will be discussed as part of the narrative analysis. We will not exclude any articles from analysis due to low quality scores, but sensitivity analysis will be

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performed to determine any effect of including studies that score less than "good" according to Agency for Health Research and Quality (AHRQ) standards.[37]

Evidence of publication bias and small-study effects will be assessed using two methods (available in the Stata meta suite).[38] First, we will examine contour-enhanced funnel plots for asymmetry using the Egger regression test.[39] Second, we will use trim-and-fill analysis to estimate effects sizes that are adjusted for publication bias.[40]

Analysis

Data that meet all inclusion criteria will first be summarised descriptively and then analysed statistically. Data analysis will be conducted using Airtable and Stata16.[34 38]

Following the format of the original review and drawing on the SWiM[41] reporting items, a narrative synthesis of study characteristics and findings will be conducted. This will provide a description and rationale for the reporting of groups used in the synthesis such as study populations, outcomes, study designs, methods used to assess the certainty of the evidence and limitations of the review. Study characteristics will be presented in summary tables across key variables (including study design, year, setting, country; population characteristics; exposure measures including number of items and whether validated; definition of racism used. If meta-analysis is not possible the nature of the relationship between exposure and outcome (positive, negative or null) across key study characteristics will be summarised, following the approach in the original review.[1]

Although meta-analysis is planned this will only become apparent when extracted data are reviewed for feasibility. If data is available we will conduct analyses of associations between racism and health for different health outcome measures, and at different time points. If possible, we will use random-effects models to aggregate effect sizes. Subgroup analyses will be conducted for age, gender and ethnicity if possible. To assess the heterogeneity of studies, we will use the Q-statistic test and the I^2 statistic. If the test for heterogeneity denoted as I^2 (if $I^2 \leq 25\%$), studies will be considered homogeneous. Based on the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) framework,[42] we will rate the quality of the overall evidence across each outcome to conclude.

Patient and Public Involvement

No patient involvement.

DISCUSSION

As this is an updated systematic review and meta-analysis, we expect that whilst there has been a significant amount of recent research conducted in this space, we do not anticipate our findings to be vastly different from our original review. This review and meta-analysis will incorporate studies with participants from all ethnic/racial/cultural backgrounds, and

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studies will not be limited to any one country or geographic area, and in doing so we anticipate to show that this is a problem facing not just one specific population but children globally. That is, we expect the review to show that racism and racial discrimination negatively impact on multiple health outcomes in children and youth from different ethnic/racial/cultural backgrounds and across contexts. We expect an increase of research in outcomes not considered in the original review, including sleep and inflammatory and immune biomarkers, as well as markers of epigenetic risk and cellular aging and of endocrine and hormonal function. Increased attention on younger age groups, vicarious as well as direct exposure, longitudinal associations, and populations and settings outside of the United States of America are also anticipated.

Due to the expectant increase in research surrounding this topic, a key contribution of the current study is to conduct a meta-analysis, which was not able to be conducted before. We expect that through this meta-analysis we will be able to show rigorous and robust evidence showing the relationship between experiences of racism and health and wellbeing outcomes for children and youth. As this is the first meta-analysis of these studies, it will provide an evidence base for future research exploring the effect of racism and child health, as well as for policy development and service delivery.

Review findings will provide essential information for future research, policy priorities and inform the development effective evidenced-based targets for interventions to ameliorate the harmful impacts of racism on child and youth health.

ETHICS AND DISSEMINATION

Ethics approval is not required as this is a review of existing empirical findings and no primary data will be collected throughout the research.

The results from this review will be disseminated in peer-review publications, conference presentations as well as communicated more broadly through factsheets and summaries disseminated through academic institution press release and policy and practice partners. Specific dissemination strategies will be co-designed with policy, practice and community stakeholders towards completion of the review.

Progress on this systematic review and meta-analysis will be updated on the International Prospective Register of Systematic Reviews (PROSPERO) (PROSPERO ID CRD42020184055) to maintain transparency.

AUTHOR CONTRIBUTIONS

NP conceptualised the review, contributed to all aspects of the protocol and the guarantor for the review. KD co-drafted the protocol, developed the search strategy with medical librarians. RP and JG assisted with drafting and revising the protocol. MT, BT, SK, YK and YP reviewed the protocol draft.

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ROLE OF THE FUNDER

The funding sources had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

COMPETING INTERESTS

The authors have no conflicts of interest relevant to this article to disclose.

REGISTRATION

The systematic review will be registered with PROSPERO, ID number CRD42020184055.

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Supplementary File 1- PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Checklist item	Page number
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a. Identify the report as a protocol of a systematic review	Title page
Update	1b. If the protocol is for an update of a previous systematic review, identify as such	Title page
Registration	2. If registered, provide the name of the registry (such as PROSPERO) and registration number	PROSPERO (CRD42020184055)
Authors:		
Contact	3a. Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title page
Contributions	3b. Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments	4. If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Not applicable
Support:		
Sources	5a. Indicate sources of financial or other support for the review	11
Sponsor	5b. Provide name for the review funder and/or sponsor	11
Role of sponsor or funder	5c. Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	11
INTRODUCTION		
Rationale	6. Describe the rationale for the review in the context of what is already known	2-3
Objectives	7. Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
METHODS		
Eligibility criteria	8. Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4-5
Information sources	9. Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6
Search strategy	10. Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6; Supplementary file 2
Study records:		
Data management	11a. Describe the mechanism(s) that will be used to manage records and data throughout the review	6-7

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Selection process	11b. State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6-7
Data collection process	11c. Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7
Data items	12. List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7
Outcomes and prioritization	13. List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	14. Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	15a. Describe criteria under which study data will be quantitatively synthesised	8
	15b. If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	8
	15c. Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	8
	15d. If quantitative synthesis is not appropriate, describe the type of summary planned	8
Meta-bias(es)	16. Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8
Confidence in cumulative evidence	17. Describe how the strength of the body of evidence will be assessed (such as GRADE)	10

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Supplementary File 2- Search Strategy

Search date: 18/7/2020

Database(s): Ovid MEDLINE(R) ALL 1946 - October 2020

Search Strategy:

1	Prejudice/ or Racism/
2	(racism or racial-discriminat* or racial-prejudice or racist-event* or racist-episode* or racial-stereotype* or race-related-stress).tw,kf.
3	((discriminat* or bias* or prejudic* or hostil* or harass* or bully* or cyberbull* or cyber-bull* or (unfair* adj1 treat*) or oppress*) adj3 (race or racial* or ethnic* or cultur* or religio* or migrant* or refugee* or asylum)).tw,kf.
4	(newborn* or new-born* or baby or babies or neonat* or neo-nat* or infan* or toddler* or pre-schooler* or preschooler* or kinder or kinders or kindergarten* or boy or boys or girl or girls or child or children or childhood or pediatric* or paediatric* or adolescen* or youth or youths or teen or teens or teenage* or school-age* or schoolage* or school-child* or schoolchild* or school-girl* or schoolgirl* or school-boy* or schoolboy* or young-person* or young-people).af.
5	Child Welfare/ or pediatric obesity/et, ep, pc
6	(Prejudice/ or *Racism/ or 2 or 3) and 5
7	obesity/et, ep, pc or body mass index/ or overweight/pc
8	Waist-Hip Ratio/
9	Blood Pressure/ or Biomarkers/
10	Hypertension/et, ep, pc
11	exp Cardiovascular Diseases/et, ep, pc
12	depression/et, ep, pc or anxiety/et, ep, pc
13	Mental Health/
14	Stress, Psychological/et, ep, pc
15	Sleep/
16	exp Sleep Wake Disorders/et, ep, pc
17	"Quality of Life"/
18	Resilience, Psychological/ or exp adaptation, psychological/
19	exp substance-related disorders/et, ep, pc or alcohol-related disorders/et, ep, pc
20	smoking/et, ep or exp tobacco smoking/et, ep, pc
21	Mental Disorders/et, ep, pc
22	Self Concept/
23	personal satisfaction/
24	exp suicide/et, ep, pc
25	conduct disorder/et, ep, pc or aggression/et, ep, pc
26	pregnancy outcome/
27	(health-care or healthcare or health-service* or clinic? or ill-health or wellbeing or wellbeing or disease* or illness* or bmi or body-mass-index or anthropometric* or WHR or waist-hip-ratio or hypertension or blood-pressure or cardiometabolic or cardio-metabolic or biomarker* or obese or obesity or overweight or depress* or anxiety or anxious* or mental-health or mental-disorder* or stress or distress* or suicid* or sleep or psychosis or tobacco or smoke* or smoking or drug? or alcohol* or substance-use or substance-related-disorder* or resilien* or self-esteem or self-worth or self-concept or quality-of-life or life-satisfaction or personal-satisfaction or conduct-disorder* or aggression).tw,kf.
28	((social or behavio* or emotion* or developmental* or psychological* or learning*) adj3 (difficul* or problem* or delay* or adjust*)).tw,kf.
29	(((pregnancy or birth or gestation*) and (outcome* or preterm or pre-term or premature or small-for-gestational-age)) or low-birthweight or low-birth-weight).tw,kf.
30	7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26
31	*obesity/et, ep, pc or *body mass index/ or *overweight/pc or *Waist-Hip Ratio/ or (*Blood Pressure/ or *Biomarkers/) or *Hypertension/et, ep, pc or exp *Cardiovascular Diseases/et, ep, pc or (*depression/et, ep, pc or *anxiety/et, ep, pc) or *Mental Health/ or *Stress, Psychological/et, ep, pc or *Sleep/ or exp *Sleep Wake Disorders/et, ep, pc or *"Quality of Life"/ or (*Resilience, Psychological/ or exp *adaptation, psychological/) or (exp *substance-related disorders/et, ep, pc or *alcohol-related disorders/et, ep, pc) or (*smoking/et, ep or exp *tobacco smoking/et, ep, pc) or *Mental Disorders/et, ep, pc or *Self Concept/ or *personal satisfaction/ or exp *suicide/et, ep, pc or (*conduct disorder/et, ep, pc or *aggression/et, ep, pc) or *pregnancy outcome/
32	(Prejudice/ or *Racism/ or 2 or 3) and (27 or 28 or 29 or 31) and 4

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33	6 or 32
34	limit 33 to (comment or editorial or letter)
35	**33 not 34**

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Supplementary File 3- Newcastle - Ottawa Quality Assessment Scale

NEWCASTLE - OTTAWA QUALITY ASSESSMENT SCALE
CASE CONTROL STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability.

Selection

1) Is the case definition adequate?
   a) yes, with independent validation ✵
   b) yes, eg record linkage or based on self reports
   c) no description

2) Representativeness of the cases
   a) consecutive or obviously representative series of cases ✵
   b) potential for selection biases or not stated

3) Selection of Controls
   a) community controls ✵
   b) hospital controls
   c) no description

4) Definition of Controls
   a) no history of disease (endpoint) ✵
   b) no description of source

Comparability

1) Comparability of cases and controls on the basis of the design or analysis
   a) study controls for _____ (Select the most important factor.) ✵
   b) study controls for any additional factor ✵ (This criteria could be modified to indicate specific control for a second important factor.)

Exposure

1) Ascertainment of exposure
   a) secure record (eg surgical records) ✵
   b) structured interview where blind to case/control status ✵
   c) interview not blinded to case/control status
   d) written self report or medical record only
   e) no description

2) Same method of ascertainment for cases and controls
   a) yes ✵
   b) no

3) Non-Response rate
   a) same rate for both groups ✵
   b) non respondents described
   c) rate different and no designation

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NEWCASTLE - OTTAWA QUALITY ASSESSMENT SCALE
COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort
   a) truly representative of the average _____ (describe) in the community ✸
   b) somewhat representative of the average _____ in the community ✸
   c) selected group of users eg nurses, volunteers
   d) no description of the derivation of the cohort

2) Selection of the non exposed cohort
   a) drawn from the same community as the exposed cohort ✸
   b) drawn from a different source
   c) no description of the derivation of the non exposed cohort

3) Ascertainment of exposure
   a) secure record (eg surgical records) ✸
   b) structured interview ✸
   c) written self report
   d) no description

4) Demonstration that outcome of interest was not present at start of study
   a) yes ✸
   b) no

Comparability

1) Comparability of cohorts on the basis of the design or analysis
   a) study controls for _____ (select the most important factor) ✸
   b) study controls for any additional factor ✸ (This criteria could be modified to indicate specific control for a second important factor.)

Outcome

1) Assessment of outcome
   a) independent blind assessment ✸
   b) record linkage ✸
   c) self report
   d) no description

2) Was follow-up long enough for outcomes to occur
   a) yes (select an adequate follow up period for outcome of interest) ✸
   b) no

3) Adequacy of follow up of cohorts
   a) complete follow up - all subjects accounted for ✸
   b) subjects lost to follow up unlikely to introduce bias - small number lost - > ____ % (select an adequate %) follow up, or description provided of those lost) ✸
   c) follow up rate < ____% (select an adequate %) and no description of those lost
   d) no statement

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BMJ Open

An updated systematic review and meta-analysis of studies examining the relationship between reported racism and health and wellbeing for children and youth: A protocol

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AN UPDATED SYSTEMATIC REVIEW AND META-ANALYSIS OF STUDIES EXAMINING THE RELATIONSHIP BETWEEN REPORTED RACISM AND HEALTH AND WELLBEING FOR CHILDREN AND YOUTH: A PROTOCOL

Naomi Priest[1, 2 *], Kate Doery[1, 2], Mandy Truong[3], Shuaijun Guo[2,4], Ryan Perry[1, 2] Brigid Trenerry[5], Saffron Karlsen[6], Yvonne Kelly[7], Yin Paradies[8]

1. Australian National University, Centre for Social Research and Methods, Canberra, Australia
2. Murdoch Children's Research Institue, Population Health, Melbourne, Australia
3. Monash University, School of Nursing and Midwifery, Melbourne, Australia
4. University of Melbourne, Department of Pediatrics, Melbourne, Australia.
5. Singapore University of Technology and Design, Lee Kuan Yew Centre for Innovative Cities, Singapore
6. University of Bristol, School of Sociology, Politics and International Studies, Bristol, United Kingdom
7. University College London, Department of Epidemiology and Public Health, London, United Kingdom
8. Deakin University, School of Humanities and Social Science, Melbourne, Australia

* Corresponding Author Associate Professor Naomi Priest, Australian National University, Canberra ACT 2600 Australia. Email: naomi.priest@anu.edu.au Phone: +61 2 6125 4849.

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ABSTRACT

Introduction: Racism is a critical determinant of health and health inequities for children and youth. This protocol aims to update the first systematic review conducted by Priest et al. 2013, including a meta-analysis of findings. Based on previous empirical data, it is anticipated that child and youth health will be negatively impacted by racism. Findings from this review will provide updated evidence of effect sizes across outcomes and identify moderators and mediators of relationships between racism and health.

Methods and analysis: This systematic review and meta-analysis will include studies that examine associations between experiences of racism and racial discrimination with health outcomes of children and youth aged 0-24 years. Exposure measures include self-reported or proxy reported systemic, interpersonal and intrapersonal racism. Outcome measures include general health and wellbeing, physical health, mental health, biological markers, healthcare utilisation and health behaviours. A comprehensive search of studies from the earliest time available to October 2020 will be conducted. A random-effects meta-analysis will examine the average effect of racism on a range of health outcomes. Study-level moderation will test the difference in effect sizes with regard to various sample and exposure characteristics. This review has been registered with the International Prospective Register of Systematic Reviews (PROSPERO) ID CRD42020184055.

Ethics and Dissemination: This review will provide evidence for future research within the field and help to support policy and practice development. Results will be widely disseminated to both academic and non-academic audiences

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through peer review publications, community summaries, and presentations to research, policy, practice and community audiences.

Key Words: Protocol, Systematic-Review, Meta-analysis, Racism, Child Health.

Strengths and Limitations

- This protocol aims to update the international review of racism and child health conducted by Priest et al.2013
- The updated protocol now extends to include study participants 0-24 years to recognise the importance of youth development.
- This protocol includes a broad search strategy that aims to capture varied exposure measures of racism and racial discrimination as well as health and wellbeing outcomes.
- This protocol describes a meta-analysis to be conducted exploring relationships between racism and health among children and youth.
- This systematic review has a bias towards papers published in English, meaning that studies not-published in English will not be included in this review.

INTRODUCTION

Racism and racial discrimination are widely recognised as critical determinants of health and health inequities for children and youth across populations and contexts.[1-3] Racism is a system of oppression that categorises and stratifies social groups into 'races', devalues and disadvantages those considered inferior and differentially allocates to them valued societal resources and opportunities.[4 5] Racism is expressed across multiple levels, including systemic or structural racism, embedded within society; interpersonal racism between two or more individuals and includes racially motivated assaults or abuse; and intrapersonal racism, whereby people take on negative stereotypes and beliefs about themselves.[6-8] Racism and racial discrimination operates in many forms including direct and vicarious racism (second hand racism), whereby individuals experience racism on a secondary level, witnessing or being informed of family, friends and strangers experiencing racism.[9 10] Racism profoundly and perniciously impacts indigenous and racialised peoples thoughout the world, including children and youth, for whom racism and racial discrimination is a major burden and influence on their health and development throughout life.[1 2 11]

Research on racism and health has predominantly focused on interpersonal experiences, with considerable evidence documenting negative health effects across multiple outcomes, including physical and mental health outcomes.[5 12 13] However, most of this evidence focused on adults, with far less research conducted among children and youth.

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Priest et al.[1] conducted the first international systematic review of quantitative studies on reported racial discrimination and the health and wellbeing of children and youth, including 121 studies. Since this report was published in 2013, the contribution of racism as a social determinant of health and wellbeing among children and youth has received growing attention.[3] There is increasing evidence of the impact of racism on pathophysiological processes (e.g. allostatic load and stress neurobiology) and biological markers (e.g. C-reactive protein and cortisol)[14] as well as on health behaviours such as sleep[15-17] among children and youth. The American Academy of Pediatrics recently issued a policy statement highlighting the impact that racism has on young people's health and health inequities and that addressing racism needs to be an urgent priority.[2]

A recent review of vicarious racism and child health found 30 studies published up to May 2016 compared with 10 studies in the previous 2013 review (with studies searched up to November 2011).[9] This represents a three-fold increase in studies examining vicarious racism and child health in approximately four and a half years. Additionally, our original review found that two-thirds of the included studies were published between 2005-2012.[1] In light of the growing research in the field, there is a need to review and reflect on the current evidence to inform future scholarship in this area.

This present systematic review and meta-analysis aims to update findings from the 2013 review conducted by Priest et al.[1] An updated systematic review is necessary to include new data, new methods and updated analysis.[18] In this instance an updated systematic review is necessary due to changing social policy and demographic contexts and new health priorities globally, as well as an increase in the number of recent publications in this area, including in different country and population contexts. The first systematic review identified that there were a limited number of longitudinal studies that have explored the health effects of racism on children and a need to expand research in this area, with a focus needed on the complex pathways to which child and youth health is impacted by experiences of racial discrimination.[1] Priest et al. called for an increase in high-quality longitudinal research utilising robust multidimensional measures of racial discrimination.[1] As highlighted since this review was published in 2013, there has been a large increase in the amount of research being conducted in this field.

To answer our research question "*To what extent are experiences of racism associated with health and wellbeing outcomes among children and youth compared to those who experience no or less racism?*" We will use the previous review (Priest et al., 2013) as a guide, building upon it and utilising an updated inclusion and exclusion strategy as well as expanding it to include a meta-analysis. As indicated by Garner et al.,[18] an updated systematic review can have an updated inclusion criteria whilst answering a similar question. This systematic review and meta-analysis aims to quantify the effects of racial discrimination on child and youth health; examine the key pathways by which racial discrimination influences these outcomes; and identify potential moderators and mediators between racism and health. This review will provide key

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recommendations for future research and inform the development of effective evidence-based strategies for addressing racism and ameliorating its harmful effects.

METHODS AND ANALYSIS

This systematic review and meta-analysis will follow the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)[19] with the PRISMA Protocols (PRISMA-P)[20] checklist followed for the writing of this protocol (see supplementary file 1). Progress on this systematic review and meta-analysis will be updated on the International Prospective Register of Systematic Reviews (PROSPERO) to maintain transparency (PROSPERO ID CRD42020184055).

Inclusion criteria

This report will include primary empirical studies that use quantitative methods including but not limited to cross-sectional; prospective and retrospective cohort; case–control designs; quasi-experimental studies and randomised control trials. Peer-reviewed journal articles (published or available as pre-prints), and dissertations/theses will be included. We will also include grey literature such as published reports. Studies that do not report primary empirical associations between racism and child and youth health will not be included. Editorials and commentaries will not be included unless they report primary empirical data. According to the PICO (Population, Intervention, Comparison and Outcomes) tool outlined in the Cochrane Handbook,[21] we defined our population, intervention (exposure of interest) and specific outcome measures.

Population

Participants will include children and youth aged up to 24 years from any racial/ethnic/cultural groups. The age range of participants has been updated since the previous review (which included participants up to 18 years) to include children and youth as per the United Nations definition.[22-24] This broader age range will allow for consideration of the health impacts of racism into late adolescence which is now considered by the Lancet Commission on Adolescent Health as extending to 24 years.[24]

Exposure

This review will extract childhood experiences of racism and racial discrimination exposures, synthesised across three main categories in the meta-analysis: systemic racism; interpersonal experiences including direct and vicarious or second hand experiences (for example, witnessing or hearing about racism experienced by family, friends or a group an individual belongs to); and intrapersonal racism (for example, adapting racist attitudes and/or beliefs within their world views). Both

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personal and proxy reports (by parents or caregivers) of experiences of racism will be recorded, as will the source of racism (categorised as peers, teachers, and community).

Terms used to determine the exposure include but are not limited to racism, discrimination, prejudice, harassment, bullying, stereotypes and unfair treatment where reason includes the victims' race/ethnicity/cultural background or proxy indicators such as migrant background, skin colour, language or accent. Although we recognise that religion is often highly racialised,[25] studies of religious discrimination will not be included as we consider this a related, but distinct form of discrimination. Religious discrimination is distinguished by being an assault on an individual or group's belief system and warrants independent investigation.[26]

There will be no restrictions placed on the timeframe of exposure to racism prior to the measurement. Retrospective adult population studies that report on childhood experiences of racism will be noted, but will not be included in our analysis.

Outcome measures

Studies will be considered if they measure health outcomes in children and youth. Health and wellbeing outcomes include measures of ill-health and illness as well as positive health outcomes across physical, mental and behavioural domains. As guided by previous reviews and research, [1 12 15 27-32] the following health and wellbeing outcomes will be included:

1. Pregnancy and birth outcomes (e.g. premature birth, low birth weight)
2. General health and wellbeing, life satisfaction and quality of life
3. Physical health (infectious disease and chronic conditions and markers e.g. body mass index, waist hip ratio, blood pressure, metabolic and cardiovascular disease, overweight, obesity)
4. Negative mental health (e.g. social and emotional difficulties, psychological distress, mental illness, suicide risk, self-harm, psychosis, antisocial behaviours including aggression and violence)
5. Positive mental health (e.g. self-esteem, self-worth and resilience)
6. Health behaviours (e.g. alcohol, tobacco, substance use) and sleep
7. Health care utilisation, healthcare costs, satisfaction with child health care system (use of screening tests, maternal child health care, access to health care and treatment, adherence to treatment)
8. Biological markers (e.g. inflammation and cardiometabolic markers)

Exclusion criteria

Studies reporting the effects of reported racism on other outcomes (e.g. cognitive development, education, employment) will not be included. Studies that do not attribute discrimination experiences to race or ethnicity but only report generalised discrimination or unfair treatment without attribution will not be included. Studies published in a language other than

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English will not be included. Qualitative studies or studies only reporting the prevalence of racism without identifying associations with health and wellbeing outcomes will not be included.

Data extraction and management

Search Strategy

The search strategy will be conducted in English and include studies from the earliest time available to October 2020. The search strategy will not be restricted to papers only published since the completion of the previous search strategy in 2011 as databases regularly back index studies and therefore some studies may have been missed by the original review.[1] The search will be checked against the original search results to ensure that all studies that have been back indexed are also included.

The search will be conducted in the Ovid Medline, Ovid PsycInfo, PubMed, ERIC and ProQuest (for dissertation/theses) databases. Reference lists of included studies will also be hand-searched for additional relevant studies. The authors will also search google scholar to identify papers and reports citing the previous review and will search grey literature databases including Open Grey, OpenDOAR and New York Academy of Medicine using keywords from the search strategy.

The search will be performed using a string template combining search terms relevant to our study population, exposure and outcomes. The search strategy template has been developed in consultation with medical library staff utilising the previous search strategy by Priest et al.[1] as a template. The search template to be used for Medline is included as supplementary file 2, which will be updated accordingly for each database.

Selection of studies

One member of the review team will conduct the initial search in the selected databases with the search results to be imported into Endnote X9,[33] All titles and abstracts of studies identified in the search will be independently screened for eligibility for inclusion independently by two members of the review team using Covidence[34] with any discrepancies resolved by the lead author. Duplicates and papers not in English will be deleted and noted in the PRISMA flowchart.[19] Full text studies will be assessed for final inclusion. Any discrepancies between members of the review team will be resolved by having a third member of the review team adjudicate the decision. Rationale for exclusion of studies will be noted throughout the screening process with a PRISMA flowchart[19] being used to show the full selection process of studies.

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Data Extraction

Once the full-text studies have been identified, members of the review team will extract the data using Airtable.[35] Two reviewers will independently extract data from each study, with inconsistencies and discrepancies resolved through discussion. Data from some studies may appear in multiple publications. If publications include unique combinations of exposure and outcome variables they will be extracted as distinct data sets; meaning that one study may be included in the meta-analysis multiple times as different data-sets due to its use of multiple measures of health or racism.

This review will examine the key characteristics of studies of reported racism and health and wellbeing among children and youth. Data to be extracted will include: authors; year of publication; study design (including sampling methods); definition of racism, exposure measure (s) (including tools/instruments and psychometric properties when applicable, method of administration including informant(s), content and time frames of exposure, targets and perpetrators, reactions/responses to racism, and settings in which racism is experienced); health and wellbeing outcome measures; measures of racial/cultural/ethnic background; study location (country/region); place of residence (urban/rural), sample size; participant demographics (age, racial/cultural backgrounds, gender, socio-economic status, migration status); study findings; prevalence of self-reported racism including exposure characteristics; nature of associations between self-reported racism and health and wellbeing outcomes including subgroup analysis when reported (mean, standard deviation, effect size); confounders, effect moderator and mediators of these associations; and study quality/critical appraisal.

Effect sizes such as coefficients and p-values for each health outcome will be extracted. Both unadjusted and adjusted effect sizes will be extracted when available, and covariates included in models recorded. Where an overall effect size is reported across a range of age groups, we will extract subgroup effect sizes when reported. In this case, only effect sizes for children and youth will be extracted.

Data including study characteristics, participant characteristics, and exposure and outcome characteristics will be extracted to be included in the narrative synthesis but will not be included in the meta-analysis.

Assessment of study quality and bias

Studies included in the review will be critically appraised to determine the validity of the study's findings from the known literature and to provide readers with the ability to make an informed decision on the quality of these findings. Two members of the review team will independently rate studies selected for the meta-analysis using the Newcastle-Ottawa Scale, a widely used tool for evaluating the quality of non-randomised studies on a range of criteria.[36] The Newcastle-Ottawa Scale (supplementary fiile 3) was determined to be the most appropriate instrument for quality assessment via a consideration of tools available in the Systematic Review Toolbox.[37] Any discrepancies between quality ratings will be

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resolved through discussion with the lead author. The quality ratings for each study will be provided in a table in the supplementary material and studies with scores deemed less than satisfactory will be discussed as part of the narrative analysis. We will not exclude any articles from analysis due to low quality scores, but sensitivity analysis will be performed to determine any effect of including studies that score less than "good" according to Agency for Health Research and Quality (AHRQ) standards.[38]

Evidence of publication bias and small-study effects will be assessed using two methods (available in the Stata meta suite).[39] First, we will examine contour-enhanced funnel plots for asymmetry using the Egger regression test.[40] Second, we will use trim-and-fill analysis to estimate effects sizes that are adjusted for publication bias.[41]

Analysis

Data that meet all inclusion criteria will first be summarised descriptively and then analysed statistically. Data analysis will be conducted using Airtable and Stata16.[35 39]

Following the format of the original review and drawing on the SWiM[42] reporting items, a narrative synthesis of study characteristics and findings will be conducted. This will provide a description and rationale for the reporting of groups used in the synthesis such as study populations, outcomes, study designs, methods used to assess the certainty of the evidence and limitations of the review. Study characteristics will be presented in summary tables across key variables (including study design, year, setting, country; population characteristics; exposure measures including number of items and whether validated; definition of racism used. If meta-analysis is not possible the nature of the relationship between exposure and outcome (positive, negative or null) across key study characteristics will be summarised, following the approach in the original review.[1]

Although meta-analysis is planned this will only become apparent when extracted data are reviewed for feasibility. If data is available we will conduct analyses of associations between racism and health for different health outcome measures, and at different time points. If possible, we will use random-effects models to aggregate effect sizes. Subgroup analyses will be conducted for age, gender and ethnicity if possible. To assess the heterogeneity of studies, we will use the Q-statistic test and the I^2 statistic. If the test for heterogeneity denoted as I^2 (if $I^2 \leq 25\%$), studies will be considered homogeneous. Based on the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) framework,[43] we will rate the quality of the overall evidence across each outcome to conclude.

Patient and Public Involvement

No patient involvement.

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DISCUSSION

As this is an updated systematic review and meta-analysis, we expect that whilst there has been a significant amount of recent research conducted in this space, we do not anticipate our findings to be vastly different from our original review. This review and meta-analysis will incorporate studies with participants from all ethnic/racial/cultural backgrounds, and studies will not be limited to any one country or geographic area, and in doing so we anticipate to show that this is a problem facing not just one specific population but children globally. That is, we expect the review to show that racism and racial discrimination negatively impact on multiple health outcomes in children and youth from different ethnic/racial/cultural backgrounds and across contexts. We expect an increase of research in outcomes not considered in the original review, including sleep and inflammatory and immune biomarkers, as well as markers of epigenetic risk and cellular aging and of endocrine and hormonal function. Increased attention on younger age groups, vicarious as well as direct exposure, longitudinal associations, and populations and settings outside of the United States of America are also anticipated.

Due to the expectant increase in research surrounding this topic, a key contribution of the current study is to conduct a meta-analysis, which was not able to be conducted before. We expect that through this meta-analysis we will be able to show rigorous and robust evidence showing the relationship between experiences of racism and health and wellbeing outcomes for children and youth. As this is the first meta-analysis of these studies, it will provide an evidence base for future research exploring the effect of racism and child health, as well as for policy development and service delivery.

Review findings will provide essential information for future research, policy priorities and inform the development effective evidenced-based targets for interventions to ameliorate the harmful impacts of racism on child and youth health.

ETHICS AND DISSEMINATION

Ethics approval is not required as this is a review of existing empirical findings and no primary data will be collected throughout the research.

The results from this review will be disseminated in peer-review publications, conference presentations as well as communicated more broadly through factsheets and summaries disseminated through academic institution press release and policy and practice partners. Specific dissemination strategies will be co-designed with policy, practice and community stakeholders towards completion of the review.

Progress on this systematic review and meta-analysis will be updated on the International Prospective Register of Systematic Reviews (PROSPERO) (PROSPERO ID CRD42020184055) to maintain transparency.

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AUTHOR CONTRIBUTIONS

NP conceptualised the review, contributed to all aspects of the protocol and the guarantor for the review. KD co-drafted the protocol, developed the search strategy with medical librarians. RP and SG assisted with drafting and revising the protocol. MT, BT, SK, YK and YP reviewed the protocol draft.

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ROLE OF THE FUNDER

The funding sources had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

COMPETING INTERESTS

The authors have no conflicts of interest relevant to this article to disclose.

REGISTRATION

The systematic review will be registered with PROSPERO, ID number CRD42020184055.

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Supplementary File 1- PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Checklist item	Page number
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a. Identify the report as a protocol of a systematic review	Title page
Update	1b. If the protocol is for an update of a previous systematic review, identify as such	Title page
Registration	2. If registered, provide the name of the registry (such as PROSPERO) and registration number	PROSPERO (CRD42020184055)
Authors:		
Contact	3a. Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title page
Contributions	3b. Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments	4. If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Not applicable
Support:		
Sources	5a. Indicate sources of financial or other support for the review	11
Sponsor	5b. Provide name for the review funder and/or sponsor	11
Role of sponsor or funder	5c. Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	11
INTRODUCTION		
Rationale	6. Describe the rationale for the review in the context of what is already known	2-3
Objectives	7. Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
METHODS		
Eligibility criteria	8. Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4-5
Information sources	9. Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6
Search strategy	10. Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6; Supplementary file 2
Study records:		
Data management	11a. Describe the mechanism(s) that will be used to manage records and data throughout the review	6-7

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Selection process	11b. State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6-7
Data collection process	11c. Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7
Data items	12. List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7
Outcomes and prioritization	13. List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	14. Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	15a. Describe criteria under which study data will be quantitatively synthesised	8
	15b. If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	8
	15c. Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	8
	15d. If quantitative synthesis is not appropriate, describe the type of summary planned	8
Meta-bias(es)	16. Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8
Confidence in cumulative evidence	17. Describe how the strength of the body of evidence will be assessed (such as GRADE)	10

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Supplementary File 2- Search Strategy

Search date: 18/7/2020

Database(s): Ovid MEDLINE(R) ALL 1946 - October 2020

Search Strategy:

1	Prejudice/ or Racism/
2	(racism or racial-discriminat* or racial-prejudice or racist-event* or racist-episode* or racial-stereotype* or race-related-stress).tw,kf.
3	((discriminat* or bias* or prejudic* or hostil* or harass* or bully* or cyberbull* or cyber-bull* or (unfair* adj1 treat*) or oppress*) adj3 (race or racial* or ethnic* or cultur* or religio* or migrant* or refugee* or asylum)).tw,kf.
4	(newborn* or new-born* or baby or babies or neonat* or neo-nat* or infan* or toddler* or pre-schooler* or preschooler* or kinder or kinders or kindergarten* or boy or boys or girl or girls or child or children or childhood or pediatric* or paediatric* or adolescen* or youth or youths or teen or teens or teenage* or school-age* or schoolage* or school-child* or schoolchild* or school-girl* or schoolgirl* or school-boy* or schoolboy* or young-person* or young-people).af.
5	Child Welfare/ or pediatric obesity/et, ep, pc
6	(Prejudice/ or *Racism/ or 2 or 3) and 5
7	obesity/et, ep, pc or body mass index/ or overweight/pc
8	Waist-Hip Ratio/
9	Blood Pressure/ or Biomarkers/
10	Hypertension/et, ep, pc
11	exp Cardiovascular Diseases/et, ep, pc
12	depression/et, ep, pc or anxiety/et, ep, pc
13	Mental Health/
14	Stress, Psychological/et, ep, pc
15	Sleep/
16	exp Sleep Wake Disorders/et, ep, pc
17	"Quality of Life"/
18	Resilience, Psychological/ or exp adaptation, psychological/
19	exp substance-related disorders/et, ep, pc or alcohol-related disorders/et, ep, pc
20	smoking/et, ep or exp tobacco smoking/et, ep, pc
21	Mental Disorders/et, ep, pc
22	Self Concept/
23	personal satisfaction/
24	exp suicide/et, ep, pc
25	conduct disorder/et, ep, pc or aggression/et, ep, pc
26	pregnancy outcome/
27	(health-care or healthcare or health-service* or clinic? or ill-health or wellbeing or wellbeing or disease* or illness* or bmi or body-mass-index or anthropometric* or WHR or waist-hip-ratio or hypertension or blood-pressure or cardiometabolic or cardio-metabolic or biomarker* or obese or obesity or overweight or depress* or anxiety or anxious* or mental-health or mental-disorder* or stress or distress* or suicid* or sleep or psychosis or tobacco or smoke* or smoking or drug? or alcohol* or substance-use or substance-related-disorder* or resilien* or self-esteem or self-worth or self-concept or quality-of-life or life-satisfaction or personal-satisfaction or conduct-disorder* or aggression).tw,kf.
28	((social or behavio* or emotion* or developmental* or psychological* or learning*) adj3 (difficul* or problem* or delay* or adjust*)).tw,kf.
29	(((pregnancy or birth or gestation*) and (outcome* or preterm or pre-term or premature or small-for-gestational-age)) or low-birthweight or low-birth-weight).tw,kf.
30	7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26
31	*obesity/et, ep, pc or *body mass index/ or *overweight/pc or *Waist-Hip Ratio/ or (*Blood Pressure/ or *Biomarkers/) or *Hypertension/et, ep, pc or exp *Cardiovascular Diseases/et, ep, pc or (*depression/et, ep, pc or *anxiety/et, ep, pc) or *Mental Health/ or *Stress, Psychological/et, ep, pc or *Sleep/ or exp *Sleep Wake Disorders/et, ep, pc or *"Quality of Life"/ or (*Resilience, Psychological/ or exp *adaptation, psychological/) or (exp *substance-related disorders/et, ep, pc or *alcohol-related disorders/et, ep, pc) or (*smoking/et, ep or exp *tobacco smoking/et, ep, pc) or *Mental Disorders/et, ep, pc or *Self Concept/ or *personal satisfaction/ or exp *suicide/et, ep, pc or (*conduct disorder/et, ep, pc or *aggression/et, ep, pc) or *pregnancy outcome/
32	(Prejudice/ or *Racism/ or 2 or 3) and (27 or 28 or 29 or 31) and 4

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33	6 or 32
34	limit 33 to (comment or editorial or letter)
35	**33 not 34**

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Supplementary File 3- Newcastle - Ottawa Quality Assessment Scale

NEWCASTLE - OTTAWA QUALITY ASSESSMENT SCALE
CASE CONTROL STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Exposure categories. A maximum of two stars can be given for Comparability.

Selection

1) Is the case definition adequate?
   a) yes, with independent validation ✵
   b) yes, eg record linkage or based on self reports
   c) no description

2) Representativeness of the cases
   a) consecutive or obviously representative series of cases ✵
   b) potential for selection biases or not stated

3) Selection of Controls
   a) community controls ✵
   b) hospital controls
   c) no description

4) Definition of Controls
   a) no history of disease (endpoint) ✵
   b) no description of source

Comparability

1) Comparability of cases and controls on the basis of the design or analysis
   a) study controls for _____ (Select the most important factor.) ✵
   b) study controls for any additional factor ✵ (This criteria could be modified to indicate specific control for a second important factor.)

Exposure

1) Ascertainment of exposure
   a) secure record (eg surgical records) ✵
   b) structured interview where blind to case/control status ✵
   c) interview not blinded to case/control status
   d) written self report or medical record only
   e) no description

2) Same method of ascertainment for cases and controls
   a) yes ✵
   b) no

3) Non-Response rate
   a) same rate for both groups ✵
   b) non respondents described
   c) rate different and no designation

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NEWCASTLE - OTTAWA QUALITY ASSESSMENT SCALE
COHORT STUDIES

Note: A study can be awarded a maximum of one star for each numbered item within the Selection and Outcome categories. A maximum of two stars can be given for Comparability

Selection

1) Representativeness of the exposed cohort
   a) truly representative of the average _____ (describe) in the community ✸
   b) somewhat representative of the average _____ in the community ✸
   c) selected group of users eg nurses, volunteers
   d) no description of the derivation of the cohort

2) Selection of the non exposed cohort
   a) drawn from the same community as the exposed cohort ✸
   b) drawn from a different source
   c) no description of the derivation of the non exposed cohort

3) Ascertainment of exposure
   a) secure record (eg surgical records) ✸
   b) structured interview ✸
   c) written self report
   d) no description

4) Demonstration that outcome of interest was not present at start of study
   a) yes ✸
   b) no

Comparability

1) Comparability of cohorts on the basis of the design or analysis
   a) study controls for _____ (select the most important factor) ✸
   b) study controls for any additional factor ✸ (This criteria could be modified to indicate specific control for a second important factor.)

Outcome

1) Assessment of outcome
   a) independent blind assessment ✸
   b) record linkage ✸
   c) self report
   d) no description

2) Was follow-up long enough for outcomes to occur
   a) yes (select an adequate follow up period for outcome of interest) ✸
   b) no

3) Adequacy of follow up of cohorts
   a) complete follow up - all subjects accounted for ✸
   b) subjects lost to follow up unlikely to introduce bias - small number lost - > ____ % (select an adequate %) follow up, or description provided of those lost) ✸
   c) follow up rate < ____% (select an adequate %) and no description of those lost
   d) no statement

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